Connor McShaffrey

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Personal Website

POSITIONS

University of Connecticut Health, Postdoctoral Fellow — To start Fall 2026
- Richard D. Berlin Center for Cell Analysis and Modeling: Vivarium Lab
- Focus: Compositional Modeling and Viability

Agora Institute for Integrative Biology, Co-founder and Board Member — 2026 – Present

National Science Foundation, Graduate Research Fellow — 2023 – Present
- Category: Computationally Intensive Research in the Life Sciences

EDUCATION

Indiana University Bloomington, Ph.D. in Cognitive Science — 2021 – Present
- GPA: 3.9/4.0
- Minor: Bioengineering
- Content Specialization: Dynamics and Constraints of Cognition

Vassar College, B.A. in Cognitive Science — 2017 – 2021
- GPA: 3.8/4.0
- Honors in Cognitive Science

Georgian Court University — 2016 – 2017
- GPA: 4.0/4.0
- Early college program for the humanities

DISSERTATION

A Theoretical Approach to the Viability of Agential Systems: Throughout the life sciences, we build models to address problems where we must assess whether an agent will survive in a particular environment. Despite this, we currently lack a principled way of understanding how the behavioral and physiological dynamics of modeled agents will unfold relative to their viability constraints — the constraints on existence. To remedy this, I develop *viability space decomposition*, an extension of dynamical systems theory that introduces new manifolds, partitioning single- and multi-agent state spaces into regions of qualitatively similar existential outcomes. I also lay the foundation for a bifurcation theory centered around these novel structures.
- Committee: Randall Beer (Chair), James Glazier (Minor Advisor), Ann-Sophie Barwich, and Jacob Foster

FELLOWSHIPS, AWARDS, & HONORS

Major Awards

National Science Foundation Graduate Research Fellowship — 2023 – Present
- Category: Computationally Intensive Research in the Life Sciences

Indiana University CogSci Outstanding Research Award — 2026

Diverse Intelligences Summer Institute Research Fellow — 2025
- Role: Group Leader

Sigma Xi Scientific Honor Society Full Member Induction — 2025

Sigma Xi Scientific Honor Society Associate Induction — 2021

Psi Chi Psychology Honor Society Induction — 2021

Travel Awards

Bhatt Family Travel Award 2024

IEEE Computational Intelligence Symposium Travel Grant 2022

IU Bloomington College of Arts and Sciences Travel Award 2022

PUBLICATIONS & PAPERS

Journal Articles

McShaffrey, C., Agmon, E., and Beer, R. D. (2026). "Matters of life and death in computational cell biology," *npj Systems Biology and Applications*. `https://doi.org/10.1038/s41540-026-00718-y`

de Leeuw, J.R., Andrews, J. ... **McShaffrey, C.** ... et al. (2021). "Words may jump-start meaning more than vision: A non-replication of early ERP effects in Boutonnet and Lupyan (2015)," *Collabra: Psychology*, 7(1), 29763. `https://doi.org/10.1525/collabra.29763`

Preprints

McShaffrey, C. and Beer, R. D. (2026). "Viability Space Decomposition: A geometric partition of survival outcomes in single- and multi-agent systems," submitted to *PRX Life*. `https://doi.org/10.48550/arXiv.2605.16753`.

Forbes, E. and **McShaffrey, C.** (2026). "Analyzing minimum viable populations in deterministic community models using viability space decomposition," submitted to *Ecology*. `https://doi.org/10.64898/2026.05.19.726018`.

Conference Proceedings

McShaffrey, C. and Beer, R. D. (2025). "Shaking up viability space: Stochasticity and survival in the transient," in *ALIFE 2025: Proceedings of the 2025 Artificial Life Conference*, MIT Press. `https://doi.org/10.1162/ISAL.a.898`

McShaffrey, C. and Beer, R. D. (2024). "Dissecting viability in multi-agent systems," in *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*, MIT Press. `https://doi.org/10.1162/isal_a_00740`

Beer, R. D., **McShaffrey, C.**, and Gaul, T.M. (2024). "Deriving the intrinsic viability constraint of an emergent individual from first principles," in *ALIFE 2024: Proceedings of the 2024 Artificial Life Conference*, MIT Press. `https://doi.org/10.1162/isal_a_00739`

McShaffrey, C. and Beer, R. D. (2023). "Decomposing viability space," in *ALIFE 2023: Proceedings of the 2023 Artificial Life Conference*, MIT Press. `https://doi.org/10.1162/isal_a_00652`

McShaffrey, C. and Beer, R. D. (2022). "Maintaining viability with multiple needs," in *2022 IEEE Symposium Series on Computational Intelligence (SSCI)*, pp. 1523–1528, IEEE. `https://doi.org/10.1109/SSCI51031.2022.10022260`

Books

Longo, E., **McShaffrey, C.**, A. Alexander, et al. (2021). *Global trends in orthodox and traditional treatment of selected infectious and non-infectious diseases*. BP International.

Papers In Preparation

McShaffrey, C. and Beer, R. D. (In Preparation). "Bifurcations in viability space: Topological transitions in dynamical systems with existential constraints," to be submitted to *SIAM Journal on Life Sciences*.

Beer, R. D., **McShaffrey, C.**, and Gaul, T.M. (In Preparation). "The intrinsic viability constraint of an emergent individual."

Abstracts & Other Publications

Claros, A. and **McShaffrey, C.** (2022). "Noisy chemotaxis parameter sweep," *NanoHub*. `https://doi.org/10.21981/68XF-XS50`

McShaffrey, C., Forbes, E., and Long, J.H. (2021). "Behavior of the encapsulated embryos of little skates, *Leucoraja erinacea*." In *Annual Meeting for the Society of Integrative and Comparative Biology*.

PRESENTATIONS

Invited Talks

Playing Dice with the Reaper: How Viability Boundaries Become "Fuzzy", Connor McShaffrey and Randall D. Beer, TONAL: Thriving On Noise in Artificial Life, Workshop at Artificial Life Conference, Japan. — 2025

Foundations of Viability, Connor McShaffrey and Randall D. Beer, Workshop on Theoretical and Experimental Approaches to Goal-Directed Behavior, Spain. — 2024

Contributed Talks

Viability Space Decomposition: A geometric partition of fate outcomes, Connor McShaffrey and Randall D. Beer, Cell State Transitions and Fate (in)decisions Workshop, National Institute for Theory and Mathematics in Biology. — 2026, Upcoming

Shaking Up Viability Space: Stochasticity and Survival in the Transient, Connor McShaffrey and Randall D. Beer, Artificial Life Conference, Japan (Virtual). `https://www.youtube.com/watch?v=fgFjWNuqqWc` — 2025

No Brain, No Problem: Processing (and Habituation) in *Stentor coeruleus*, Reshma Joy, Temple Maduoma, Connor McShaffrey, and Zelalem Taye, Diverse Intelligences Summer Institute, Scotland. — 2025

Dissecting Viability in Multi-Agent Systems, Connor McShaffrey and Randall D. Beer, Artificial Life Conference, Denmark. — 2024

The Global Structure of Viability Space, Connor McShaffrey and Randall D. Beer, Artificial Life Conference, Denmark. `https://www.youtube.com/watch?v=koBc9KRDUy8` — 2024

Deriving the Intrinsic Viability Constraint of an Emergent Individual from First Principles, Randall D. Beer, Connor McShaffrey, and Thomas M. Gaul, Artificial Life Conference, Denmark. `https://www.youtube.com/watch?v=R93pRjfwsJ4` — 2024

Decomposing Viability Space, Connor McShaffrey and Randall D. Beer, Artificial Life Conference, Japan. `https://www.youtube.com/watch?v=58f0Rxv4DMw` — 2023

Decomposing Viability Space, Connor McShaffrey and Randall D. Beer, Society for Mathematical Biology Annual Meeting, Ohio. — 2023

Maintaining Viability with Multiple Needs, Connor McShaffrey and Randall D. Beer, IEEE Symposium Series on Computational Intelligence: Artificial Life, Singapore. — 2022

Replication Rumbles: Progress and Hindsight Working with Computational Models, Andrew Claros and Connor McShaffrey, NSF nanoBIO Education Showcase, Virtual. — 2021

Behavior of the Encapsulated Embryos of Little Skates *Leucoraja erinacea*, Connor McShaffrey, Eden Forbes, and John Long, Society for Integrative and Comparative Biology Annual Meeting, Virtual. — 2021

Departmental Talks

Organizing Principles for Viability Space, Connor McShaffrey and Randall D. Beer, IU Bloomington CogLunch. — 2024

Posters

Translating Between Ordinary Differential Equation and Agent-Based Models, William Lin, Onaghise Omoregbe, and Connor McShaffrey, Luddy Undergraduate Research Symposium, Indiana University Bloomington. — 2024

Modeling the Ventilatory Behavior of Embryos of Little Skates, Connor McShaffrey and John Long, Undergraduate Research Summer Institute Symposium, Vassar College. 2020

To Err is Human: How we (Mis)Assemble Robots, Xinyue Hue, Connor McShaffrey, Candido Diaz, and John Long, Undergraduate Research Summer Institute Symposium, Vassar College. 2019

TEACHING & MENTORSHIP

Undergraduate Advising

Undergraduate Honors Thesis Mentor: *Biological Organisation and Viability: A Theory of Autopoiesis in Cellular and Euclidean Automata*, Thomas M. Gaul. Link to Thesis. 2024 - 2025

Project Mentor: *Autopoiesis in RealLife Euclidean Automata*, Thomas M. Gaul. Link to Paper. 2023 - 2024

Project Mentor: *Deriving The Intrinsic Viability Constraint of an Emergent Individual from First Principles*, Thomas M. Gaul. Link to Paper. 2023 - 2024

Project Mentor: *Translating Between Ordinary Differential Equation and Agent-Based Models*, William Lin and Onaghise Omoregbe. 2024

Taught Courses

Assistant Instructor for Computation in the Cognitive and Information Sciences (Indiana University Bloomington, COGSQ-320): "Develop computer programming skills, learn to write programs that simulate cognitive processes, and run experiments with human subjects. The relation between computation and intelligence and a selection of approaches from artificial intelligence will be explored." 2022

Assistant Instructor for Multicell Virtual Tissue Modeling Online Summer School and Hackathon (Indiana University Bloomington): "Mechanistic modeling is an integral part of contemporary bioscience, used for hypothesis generation and testing, experiment design and interpretation, and the design of therapeutic interventions. The CompuCell3D modeling environment allows researchers to rapidly build and execute complex Virtual Tissue simulations with minimal programming experience. CompuCell3D enables biological simulation from the subcellular scale to the tissue scale, such as tumor growth, what happens to tissues and cells when exposed to toxic compounds, viral spread in tissues, early embryonic development, intra/extra-cellular biochemical networks, and more." 2022

PROFESSIONAL SERVICE

Volunteer

Secretary: Agora Institute for Integrative Biology 2026 - Present

Panels

Graduate Student Panelist: Annual Midwest Undergraduate Cognitive Science Conference. 2022, 2023, & 2025

Graduate School Orientation Panelist 2022

ACTIVE MEMBERSHIPS

Cognitive Science Society 2017 - Present

International Society for Artificial Life 2023 - Present

Society for Mathematical Biology 2023 - Present

Society for Industrial and Applied Mathematics 2025 - Present
American Physical Society 2025 - Present

SKILLS

Certificates

Network Modeling Course Certificate, Center for Reproducible Biomedical Modeling 2022

Computational Neuroscience Course Certificate, Neuromatch Academy 2021

General

Coding: Python, Mathematica, LaTeX

Mathematics: Dynamical Systems Theory (nonlinear, hybrid, stochastic), Mathematical/Computational Biology, Complex Systems

Misc: Teaching/mentoring, lab/team management, animal husbandry, agent-based models, idealized models, evolutionary algorithms, microscopy